BELLABEAT FITNESS TRACKER

Wellness Device Usage & Product Strategy Analysis

Case Study Report

Google BigQuery	Python / Colab	Power BI	33 Users

Analyst	Period	Submitted
Brojo Mohan Dutta	FitBit tracker logs (~2-month window)	2026

1. Introduction

This report presents the findings of a full-stack data analysis conducted on FitBit smart-device tracker data, used as a proxy for how consumers currently engage with wellness wearables. The analysis was completed as a capstone-style portfolio project following the industry-standard Ask → Prepare → Process → Analyze → Share → Act framework.

Business Context

- Bellabeat is a high-tech company that manufactures health-focused smart products for women, including the Leaf, Time, and Spring devices and the Bellabeat app.
- Bellabeat's co-founder and Chief Creative Officer, Urška Sršen, believes analyzing existing smart-device fitness data can reveal untapped opportunities for growth.
- The marketing team wants to understand how consumers already use non-Bellabeat smart devices, to apply those insights to Bellabeat's own product and marketing strategy.
- Because no Bellabeat-proprietary usage dataset was available for this analysis, publicly available FitBit tracker data is used as an industry proxy for smart-device behavior.

Business Question

Primary Question Assigned to This Analysis	How do consumers use smart fitness devices day-to-day, and what does that suggest for Bellabeat's product and marketing strategy?

Secondary questions guiding the analysis:

- What are the trends in smart-device usage (activity, sleep, weight)?
- How could these trends apply to Bellabeat customers?
- How could these trends influence Bellabeat's marketing strategy?

2. Ask

Business Task Statement

Formal Business Task	Analyze FitBit smart-device tracker data across activity, sleep, and weight logging to identify behavioral usage patterns, and produce actionable, data-backed product and marketing recommendations for the Bellabeat app.

Key Stakeholders

Stakeholder	Role	Interest in This Analysis
Urška Sršen	Co-founder & Chief Creative Officer	Needs data-driven insight to guide product/marketing direction
Sando Mur	Co-founder & Mathematician, Executive Team	Must approve recommendations — requires compelling evidence
Bellabeat Marketing Analytics Team	Data team	Will implement and track campaigns based on findings

3. Prepare

Data Source

The dataset used is the FitBit Fitness Tracker Data, made publicly available on Kaggle by Möbius under a CC0 Public Domain license, originally gathered via an Amazon Mechanical Turk survey of consenting FitBit users.

Attribute	Detail
Source	FitBit Fitness Tracker Data — Kaggle (Möbius)
License	CC0: Public Domain
Period	~2-month tracking window (exact survey dates not provided by source)
Format	6 relational CSV tables loaded into BigQuery
Total users	33 unique users
Storage	Google BigQuery (fitness_tracking_analysis dataset)
Privacy	Anonymized user_id; no PII

Data Credibility — ROCCC Assessment

Criterion	Assessment	Notes
Reliable	⚠ Partial	Only 33 users; self-selected Mechanical Turk sample, not verified against a larger population
Original	⚠ Partial	Third-party survey data, not collected first-party by Bellabeat
Comprehensive	✕ Limited	No demographic fields (age, gender, location); sleep data covers 24/33 users, weight only 8/33
Current	⚠ Unclear	Exact collection year not published by the source — treat trends as directional, not time-current
Cited	✓ Yes	Public CC0-licensed Kaggle dataset with clear attribution

Limitations

- Only 33 users — too small a sample for statistically robust population-level claims
- Short ~2-month observation window — no visibility into seasonal patterns
- No demographic data — segmentation is behavior-only (steps-based), not age/gender-informed
- Sleep (24/33) and weight (8/33) logging is opt-in, producing large data-completeness gaps vs. passively-collected steps/calories

4. Process

Tools Selected

Tool	Purpose	Justification
Google BigQuery	Data cleaning, joins, transformation, analysis queries	Relational joins across 6 tables (activity, sleep, weight, hourly steps/intensity, heart rate) at cloud-SQL scale with SQL query history for reproducibility
Python / Google Colab	Statistical testing, correlation, visualization	SciPy Pearson correlation and Welch's t-test require Python; Colab needs no local setup
Power BI	Interactive dashboards for stakeholder presentation	Industry-standard BI tool; supports slicers, gauges, and multi-page dashboard navigation

Data Pipeline

The following 10-step SQL pipeline was executed in Google BigQuery:

Step	Script	Output	Key Action
1	01_dataset_setup_and_validation.sql	Validation only	Confirm dataset + all 6 raw source tables exist
2	02_clean_daily_activity.sql	cleaned_daily_activity (~940 rows)	Cast types, add day_of_week + total_active_minutes + data_source tag
3	03_clean_sleep_data.sql	cleaned_sleep_data (~413 rows)	Add sleep_efficiency_pct derived metric
4	04_clean_hourly_steps.sql	cleaned_hourly_steps (24,568 rows)	PARSE_TIME string→TIME conversion, hour extraction
5	05_clean_hourly_intensities.sql	cleaned_hourly_intensities	PARSE_TIME conversion, hour extraction
6	06_clean_weight_log.sql	cleaned_weight_log (~67 rows)	SAFE_CAST on Fat column, BMI category bucketing
7	07_clean_heart_rate.sql	cleaned_heart_rate + hourly_heart_rate	Filter to 30–220 bpm, roll seconds up to hourly avg
8	08_master_user_activity.sql	master_user_activity	LEFT JOIN activity+sleep+weight; add activity_score + daily_activity_segment
9	09_analysis_queries.sql	8 result tables (A1–A8)	Answered all business questions
10	10_export_tables.sql	7 CSVs (GCS export)	Optimized exports for Python/Power BI

Calculated Columns Added

Column	Formula	Purpose
day_of_week	FORMAT_DATE('%A', activity_date)	Weekly pattern analysis

Column	Formula	Purpose
total_active_minutes	very + fairly + lightly active minutes	Single activity-volume metric
sleep_efficiency_pct	(minutes_asleep / time_in_bed) × 100	Sleep quality metric
daily_activity_segment	CASE on total_steps thresholds (5k/7.5k/10k)	4-tier behavioral segmentation
activity_score	Weighted composite of steps, very-active min, total-active min (0–100)	Single comparable engagement score
bmi_category	CASE on BMI thresholds	Underweight/Normal/Overweight/Obese bucketing
hour_of_day	EXTRACT(HOUR FROM PARSE_TIME(...))	Hourly engagement-timing analysis

Cleaning Filters Applied

Issue	Filter / Fix Applied	Reason
Whitespace risk on IDs	TRIM(CAST(Id AS STRING))	Prevent join mismatches across tables
Fat column NULL-cast failure	SAFE_CAST(Fat AS INT64)	Preserve NULLs instead of erroring the query
Time stored as STRING	PARSE_TIME('%I:%M:%S %p', ...)	Enable proper hour-of-day extraction
Impossible heart-rate values	Value BETWEEN 30 AND 220	Remove physiologically invalid sensor noise
Duplicate rows	SELECT DISTINCT on all cleaning steps	Remove system duplicate records
No data provenance	data_source = 'fitbit_kaggle' tag added	Traceability if additional sources are merged later

Post-Cleaning Validation Results

- cleaned_daily_activity: ~940 rows, 33 unique users
- cleaned_sleep_data: ~413 rows, 24 unique users
- cleaned_hourly_steps: 24,568 rows, hours 0–23 confirmed
- cleaned_weight_log: ~67 rows, 8 unique users

Validation SELECT COUNT(*) blocks run after every CREATE OR REPLACE TABLE — see individual SQL scripts.

5. Analyze

⚠ The Python notebook (bellabeat_analysis.ipynb) is built through Cell 09 / Chart 6 of 13. Cells 10–19 (Charts 7–13) and full statistical outputs (exact r-values, t-statistics, means) are pending execution against real exported CSVs — figures below marked TBD should be replaced once the notebook is run end-to-end.

5.1 User Segmentation Overview

Metric	Value
Total users	33
Users with sleep data	24 (72.7%)

Metric	Value
Users with weight data	8 (24.2%)
Sedentary-leaning users (<5,000 steps/day)	≈ 30% of users
Segmentation tiers	Sedentary / Lightly Active / Fairly Active / Very Active

5.2 Statistical Validation (pending notebook execution)

Test	Purpose	Status
Pearson correlation + OLS regression	Steps vs. calories relationship strength	TBD — Cell 07
Welch's t-test	Calories burned, active vs. sedentary segments	TBD — Cell 08
Correlation matrix (12 variables)	Cross-metric relationship map	TBD — Cell 06

Directionally, steps and calories are positively associated (consistent with expected physiological effort-to-burn relationship); exact r/R^2 pending execution.

5.3 Temporal Patterns

- Activity peaks in early evening hours, per hourly steps analysis (Query A3 / Cell 13) — exact peak hour TBD pending chart execution.
- Day-of-week trends available via day_of_week_trends.csv (Query A2 / Cell 12) — TBD pending execution.

5.4 Sleep Quality

Metric	Status
Users tracking sleep	24 / 33 (72.7%)
Avg sleep efficiency % (time asleep ÷ time in bed)	TBD — Query A5 / Cell 14
Nights under 7 hrs asleep	TBD — Query A5 / Cell 14
Qualitative signal	Sleep-tracking users show more consistent overall engagement than non-trackers

5.5 Sedentary Time Distribution

Query A6 buckets days into Low (<10 hrs sedentary), Medium (10–15 hrs), and High (>15 hrs) sedentary time. Approximately 30% of users average under 5,000 steps/day combined with 15+ sedentary hours — the segment targeted by Recommendation 2 below.

6. Share

Power BI Dashboard Summary

Interactive Power BI dashboards were developed from the 9 pbi_ export tables, to communicate findings to different audiences within Bellabeat:

Dashboard	Audience	Core Message	Key Visuals
D1 — Engagement Overview	Bellabeat leadership	Scale of the sedentary-user problem and segment mix	Donut segment chart, 3 day-of-week bar charts, scatter chart
D2 — Sleep & Wellness	Product / membership team	Sleep tracking as a retention lever	User slicer, sleep gauge charts
D3 — Engagement Timing	Marketing / app notifications team	When to reach users for max response	Activity-score gauge, hourly area chart, weekly trend line, notification-timing bar chart

Source export tables: pbi_master_activity.csv, pbi_user_summary.csv, pbi_user_segments.csv, pbi_day_trends.csv, pbi_peak_hours.csv, pbi_scatter.csv, pbi_sleep.csv, pbi_weekly_trend.csv, pbi_hourly_combined.csv (see notebook Cell 18).

7. Act — Top 3 Recommendations

Recommendation 1 — Target the 5–7pm Engagement Window

- Evidence: Activity peaks sharply in the early evening across the hourly steps data (Query A3).
- Action: Schedule motivational app notifications, step challenges, and hydration reminders in the 5–7pm window rather than the morning, when users are least active.
- Supporting metric: Peak hour from hourly_steps analysis (cleaned_hourly_steps, Chart 7 pending).

Recommendation 2 — Re-engage Sedentary Users with a "Move More" Feature

- Evidence: ≈30% of users average fewer than 5,000 steps/day and spend 15+ hours sedentary (Query A6, sedentary_distribution).
- Action: Add a gentle movement-nudge feature to the Bellabeat app — e.g. a 10-minute walk prompt after 2 hours of inactivity — to convert this underserved segment into consistent, engaged users.
- Supporting metric: Sedentary distribution + daily_activity_segment mix from master_user_activity.

Recommendation 3 — Market Sleep Coaching as a Premium Differentiator

- Evidence: Only 24 of 33 users tracked sleep, yet those who did showed more consistent overall engagement.
- Action: Feature personalized sleep coaching (time-to-sleep optimization, wake-window alerts) prominently in Bellabeat Membership as a key upgrade driver — sleep data is deeply personal and tends to drive high retention.
- Supporting metric: sleep_efficiency_pct + days-tracked comparison (Query A5, sleep_analysis).

8. Additional Data for Future Analysis

The following data sources would significantly expand the quality of recommendations if made available:

- Demographic data — age, gender, location — to identify which Bellabeat customer segments most resemble the sedentary-leaning or sleep-engaged FitBit cohorts
- Longer time window — a full 12-month dataset to test seasonal effects on activity and sleep
- Bellabeat-proprietary app usage data — in-app notification response rates to directly validate the 5–7pm timing recommendation
- Membership conversion history — to link engagement patterns (sleep tracking, activity score) to actual upgrade behavior
- Larger sample size — 33 users is too small to generalize confidently to Bellabeat's full customer base

9. Methodology & Reproducibility

Phase	Tool	Script / File	Output
Setup & validation	Google BigQuery	sql/01_dataset_setup_and_validation.sql	Confirms dataset + 6 raw tables exist
Cleaning (6 tables)	Google BigQuery	sql/02–07_clean_*.sql	6 cleaned tables + hourly_heart_rate rollup
Master join	Google BigQuery	sql/08_master_user_activity.sql	master_user_activity (primary analysis table)
Analysis queries	Google BigQuery	sql/09_analysis_queries.sql	8 result tables (A1–A8)
Export	Google BigQuery	sql/10_export_tables.sql	7 CSVs to GCS / local
Statistical testing & EDA	Python / Colab	python/bellabeat_analysis.ipynb (+ .py)	Charts 1–6 built; Charts 7–13 pending
Visualization	Power BI	powerbi/ (.pbix)	3 interactive dashboards (9 export CSVs)

All SQL scripts, the Python notebook, and documentation are organized in this GitHub repository. To reproduce: run SQL scripts 01→10 in order in BigQuery, upload the 7 exported CSVs to Colab, run the notebook top to bottom, then load the 9 pbi_ CSVs into Power BI.